

NOTES ON CATEGORICAL ENUMERATIVE INVARIANTS

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ABSTRACT. The goal is to present Caldararu-Tu's construction of categorical enumerative invariants by introducing some basic definitions and sketching the proof strategy.

1. NON-COMMUTATIVE PRELIMINARIES

In this section we use homological degree conventions. We leave precise signs aside for the moment.

1.1. Cyclic A-infinity algebras. A shifted A_∞ -algebra is a differential graded $\mathbb{K}$ -vector space A endowed with multilinear maps

$$m_n : A^{\otimes n} \rightarrow A, \quad n \geq 2,$$

of degree $n - 2$, which satisfy the A_∞ -relations

$$\partial(m_n) = \sum_{\substack{p \geq 0, q \geq 2, r \geq 1 \\ p+q+r=n}} \pm m_{p+1+r}(\text{id}^{\otimes p} \otimes m_q \otimes \text{id}^{\otimes r}).$$

Here, the boundary of a multilinear map is defined by $\partial(f) := d_A f - (-1)^{\deg f} f d_{A^{\otimes n}}$.

An A_∞ -algebra A is *cyclic* if it is endowed with a non-degenerate graded symmetric bilinear form $\langle -, - \rangle : A \otimes A \rightarrow \mathbb{K}$ of fixed degree d , such that for all $a_0, \dots, a_n \in A$ we have

$$\langle m_n(a_0, \dots, a_{n-1}), a_n \rangle = \pm \langle m_n(a_1, \dots, a_n), a_0 \rangle.$$

The degree d of the pairing is called the *Calabi–Yau dimension* of A .

1.2. Hochschild chains. We denote by $L := CH_\bullet(A)$ the *Hochschild chain complex* of A , i.e. the complex $\bigoplus_{n \geq 0} A \otimes A^{\otimes n}$ with differential $b : L \rightarrow L$ of degree -1 given by the formula

$$\begin{aligned} b(a_0 \otimes \dots \otimes a_n) &:= \sum_{k \geq 1} \sum_{i=0}^{n-k+1} \pm a_0 \otimes \dots \otimes a_{i-1} \otimes m_k(a_i, \dots, a_{i+k-1}) \otimes a_{i+k} \otimes \dots \otimes a_n \\ &+ \sum_{k \geq 1} \sum_{i=n-k+2}^n \pm m_k(a_i, \dots, a_n, a_0, \dots, a_{i+k-n-2}) \otimes a_{i+k-n-1} \otimes \dots \otimes a_{i-1}. \end{aligned}$$

We have a circle action given by Connes operator $B : L \rightarrow L$, of degree $+1$, via the formula:

$$B(a_0 \otimes \dots \otimes a_n) := \sum_{i=0}^n \pm 1 \otimes a_i \otimes a_{i+1} \otimes \dots \otimes a_n \otimes a_0 \otimes \dots \otimes a_{i-1}.$$

The two satisfy the relations

$$(1.1) \quad b^2 = 0, \quad B^2 = 0, \quad (b + B)^2 = 0.$$

Adding a formal variable u of degree -2 , we can then form the *periodic cyclic*, *negative cyclic* and *cyclic* mixed graded complexes

$$\begin{aligned} L^{\text{Tate}} &:= (L((u)), b + uB), \\ L_+ &:= (L[[u]], b + uB), \\ L_- &:= (L[u^{-1}], b + uB) = L^{\text{Tate}} / L_+. \end{aligned}$$

The natural u -adic filtration induces the *non-commutative Hodge filtration* in homology $H_\bullet(L_+)$. We say that A has the *Hodge-de Rham degeneration property* if the associated spectral sequence collapses at the first page.

1.3. Splitting of the Hodge filtration. Using the pairing and A_∞ operations of A [9, Prop. 5.22], one defines the (chain-level) *Mukai pairing*

$$\langle -, - \rangle_{\text{Muk}} : L \otimes L \rightarrow \mathbb{K},$$

with respect to which the circle operator B is self-adjoint. This allows one to define the *higher residue pairing*

$$\langle -, - \rangle_{\text{hres}} : L^{\text{Tate}} \otimes L^{\text{Tate}} \rightarrow \mathbb{K}((u))$$

via the formula

$$\left\langle \sum_k x_k u^k, \sum_l y_l u^l \right\rangle_{\text{hres}} := \sum_{k,l} (-1)^l \langle x_k, y_l \rangle_{\text{Muk}} u^{k+l}.$$

Since B is self-adjoint with respect to the Mukai pairing, both pairings descend to homology.

Definition 1. A *splitting* of the non-commutative Hodge filtration of A is a graded map of $\mathbb{Z}/2\mathbb{Z}$ -graded vector spaces

$$s : H_\bullet(L) \rightarrow H_\bullet(L_+)$$

such that

- (1) it splits the canonical projection $H_\bullet(L_+) \rightarrow H_\bullet(L)$, and
- (2) for any a, b in $H_\bullet(L)$, we have $\langle a, b \rangle_{\text{Muk}} = \langle s(a), s(b) \rangle_{\text{hres}}$.

2. DEFINITION OF CATEGORICAL ENUMERATIVE INVARIANTS

In this section, we let A be a $\mathbb{Z}/2\mathbb{Z}$ -graded cyclic A_∞ -algebra over a field of characteristic zero, and we suppose further that A is smooth, finite dimensional, unital, and satisfies the Hodge-de Rham degeneration property.

Let $\mathcal{M}_{g,k,l}$ denote the moduli space of Riemann surfaces of genus g with $k+l$ marked points, k labeled as inputs and l labeled as outputs. In order to define categorical enumerative invariants, Caldararu–Tu consider the ribbon graphs model $M_{g,k,l}$ for $C_\bullet(\mathcal{M}_{g,k,l})$, see [10, 8, 6, 7, 2]. This is a dg-prop, whose totalisation is a Lie algebra¹

$$(2.1) \quad \mathfrak{g} := \text{Tot}(M_{g,k,l}).$$

Theorem 1 ([3]). *There is a canonical element $v \in \mathfrak{g}$ which satisfies the Maurer–Cartan equation*

$$dv + \frac{1}{2}[v, v] = 0,$$

and is unique up to homotopy (i.e. gauge equivalence between Maurer–Cartan elements).

This element is called the *string vertex*, and its genus g , k, l -points components $v_{g,k,l}$ are referred to as *string vertices*.

It is built out of the fundamental classes of the moduli spaces $\mathcal{M}_{g,k,l}$, and can be computed recursively. Indeed, according to Caldararu–Tu [1, Sec. 5.4] the Maurer–Cartan equation determines completely $v_{g,k,l}$ from previously computed string vertices; it thus suffices to fix $v_{0,1,2}$ to specify v uniquely.

Theorem 2 ([2, 10]). *The Hochschild chains of any cyclic A_∞ -algebra A form a closed TCFT. That is, we have a morphism of props $C_\bullet(\mathcal{M}_{g,k,l}) \rightarrow \text{End}_L$, or equivalently a morphism of props*

$$(2.2) \quad \rho : M_{g,k,l} \rightarrow \text{End}_L.$$

Denoting by

$$(2.3) \quad \mathfrak{h}_A := \text{Tot}(\text{End}_L),$$

the totalisation Lie algebra of End_L , we get a Lie algebra morphism

$$(2.4) \quad \mathfrak{g} \xrightarrow{\rho} \mathfrak{h}_A.$$

¹Note that I am using here notations that differ slightly from the ones in [1].

The image of the string vertex v under this morphism encodes the Gromov–Witten type invariants. In order to obtain a power series in Hochschild (or cyclic) homology classes, Caldararu–Tu use further a splitting of the non-commutative Hodge filtration of A in order to define a formality L_∞ quasi-isomorphism

$$(2.5) \quad \mathfrak{h}_A \xrightarrow{s} \mathfrak{h}_A^{\text{triv}}$$

between the $\mathfrak{h}_A$ and itself, with trivial bracket.

Definition 2. *Categorical enumerative invariants* are the genus g , n -point components $F_{g,n}^{A,s}$ of the image of the string vertex $v \in \mathfrak{g}$ under the composite $\mathfrak{g} \xrightarrow{\rho} \mathfrak{h}_A \xrightarrow{s} \mathfrak{h}_A^{\text{triv}}$.

Caldararu–Tu do not provide an explicit description of the morphism ρ , and rely on its existence, as proved in [2]. For the present purposes, only the images of the string vertices are needed, and they will be simply defined as is in Theorem 3 below. However, as shown in [10, Sec. 6.2], the action ρ can be made explicit following the Kontsevich–Soibelman recipe [7, pp. 58.62].

The L_∞ morphism s is defined as a sum-over-graphs formula (the graphs which appear below in Theorem 3, which are distinct from the graphs defining $M_{g,k,l}$ and $v_{g,k,l}$). It uses a chain-level lift of the splitting of the non-commutative Hodge filtration obtained via the homotopy transfer theorem, as well as the integration of a family of Lie algebras according to a method of Fukaya [5] and Fiorenza [4]. Note that Caldararu–Tu provide the formula for s , and prove that it is indeed a L_∞ morphism, without providing the derivation of it from Fukaya’s method.

3. AN EXPLICIT FORMULA FOR CATEGORICAL ENUMERATIVE INVARIANTS

The main contribution of the Caldararu–Tu paper is to give a sum-over-graphs formula for $F_{g,n}^{A,s}$. Our goal now is to describe it as precisely as possible in Theorem 3 below.

3.1. Graphs. Let $\text{Gra}(g, 1, n-1, m)$ denote the set of *stable partially directed graphs*, that is graphs G with n leaves (1 labeled incoming, $n-1$ labeled outgoing) and m vertices, equipped with

- a genus labeling function $g : V(G) \rightarrow \mathbb{Z}_{\geq 0}$,
- a subset of directed edges,
- a spanning tree T ,

such that

- there is no directed loop,
- each vertex has at least one incoming half-edge,
- for every vertex, the genus g and the valency η satisfy the relation $2g - 2 + \eta > 0$.

We denote by $\text{Aut}(G)$ the size of the automorphism group of G .

3.2. Graph weight. To such a graph G , Caldararu–Tu associate a rational number $w(G)$ as follows. An edge e of the spanning tree T of G is *contractable* if the graph G/e obtained by contracting all the directed edges connecting the two endpoints of e is again a stable partially directed graph. Denote by T^{cont} the set of contractable edges of T , and by E^{nl} the set of non-loop edges of G . Then, the weight $w(G)$ is defined inductively by the conditions

- if G has only one vertex, $w(G) := 1$,
- otherwise $w(G) := \frac{1}{|E^{\text{nl}}|} \sum_{e \in T^{\text{cont}}} w(G/e)$.

3.3. Graph decorations. Finally, given a cyclic A_∞ -algebra A endowed with a splitting s of the non-commutative Hodge filtration, Caldararu–Tu associate a certain linear map to any vertex (resp. edge or leg) of a stable partially directed graph G . The graph then indicates how these different “contributions” should be concatenated in order to obtain an element in $\text{Hom}(L_-, \text{Sym}^{n-1} L)$.

Using the homotopy transfer theorem, Caldararu–Tu fix an arbitrary chain-level lift S of the splitting s , as well as an inverse S^{-1} . For $\alpha = \alpha_0 + \alpha_1 u^{-1} + \dots \in L_-$, they denote by $\Theta : L_- \rightarrow L_+[1]$ the circle action map

$$\Theta(\alpha) := B(\alpha_0)$$

They then define a bounding homotopy F for Θ , using S and S^{-1} . Analogously, there is a bounding homotopy H for the circle action $\Omega : \text{Sym}^2 L_- \rightarrow \mathfrak{h}_A$,

$$\Omega(x, y) := \langle B(x_0), y_0 \rangle_{\text{Muk}}$$

defined by using the maps S and S^{-1} . Finally, there is a bounding homotopy $\delta : L_- \otimes L_- \rightarrow \mathbb{K}$ which measures the failure of H to be symmetric.

We are now ready to describe each of the contributions.

3.3.1. Vertices. A vertex v is decorated by the string vertex corresponding to its genus and valency:

$$\text{Cont}(v) := \rho(v_{g(v), k(v), l(v)})$$

It depends only on A .

3.3.2. Legs. The legs contributions depend on the splitting of the non-commutative Hodge filtration.

$$\text{Cont}(l) := \begin{cases} S & l \text{ is incoming,} \\ S^{-1} & l \text{ is outgoing.} \end{cases}$$

3.3.3. Directed edges. The directed edges contributions depend on the splitting of the non-commutative Hodge filtration, and correspond to the (homotopy) circle action.

$$\text{Cont}(e) := \begin{cases} F & e \text{ is in the spanning tree,} \\ \Theta & e \text{ is not in the spanning tree.} \end{cases}$$

3.3.4. Undirected edges. The undirected edges contributions also depend on the splitting of the non-commutative Hodge filtration.

$$\text{Cont}(e) := \begin{cases} \delta & e \text{ is in the spanning tree,} \\ H & e \text{ is not in the spanning tree.} \end{cases}$$

Let us denote by $G(v, F, \Theta, \delta, H)$ the composition of the operators performed according to the graph G , in order to produce an element in $\text{Hom}(L_-, \text{Sym}^{n-1} L) \subset \mathfrak{h}_A^{\text{triv}}$.

Theorem 3 ([1, Theorem 1.3]). *For any $g \geq 0$ and $n \geq 1$ such that $2g - 2 + n > 0$, the categorical enumerative invariant $F_{g,n}^{A,s}$ is given by the formula*

$$(3.1) \quad F_{g,n}^{A,s} = \sum_{\substack{m \geq 1 \\ G \in \text{Gra}(g, 1, n-1, m)}} (-1)^{m-1} \frac{w(G)}{\text{Aut}(G)} G(v, F, \Theta, \delta, H)$$

4. QUESTIONS

Question 1. Check the relations (1.1). Note that you can use the cyclic bar construction to check $b^2 = 0$ more conceptually.

Question 2. Specialize the formula of [9, Prop. 5.22] to the case of a cyclic A_∞ -algebra, and express the Mukai pairing purely in terms of the A_∞ operations and cyclic pairing.

Question 3. Show that the Mukai and higher residue pairing descend to cyclic homology.

Question 4. How is the formula for the L_∞ morphism derived from Fukaya [5] and Fiorenza [4] works? For the definition of partially directed graphs, Caldararu–Tu reference Getzler–Kapranov’s “Modular operads” paper. Is there a deeper connection to their work?

Question 5. Can one compute the values of the string vertices given in [1, Sec. 5.4] from the choice of value of the string vertex $v_{0,1,2}$? How is this latter value chosen in the first place?

Question 6. The *genus* of a graph G is defined to be

$$g(G) := \sum_{v \in V(G)} g(v) + \text{rank } H_1(G).$$

Show that all genus zero graphs have weight 1.

Question 7. Where does the graph weight come from, conceptually?

Question 8. Explain the origins of the names “non-commutative Hodge filtration”, “Hodge–de Rham degeneration property”, “Mukai pairing”, , and “splitting of the Hodge filtration”. In what sense are all these notions “non-commutative analogues” of classical notions?

Question 9. Where does the name “Tate complex” come from?

Solution (Bingyu). TBW

□

Question 10 (Mingyuan). Great to have Theorem 3, but how does one compute a number from it?

Solution. See Joel’s talk!

□

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